

ISE 5970: Advanced Systems Optimization

University of Oklahoma School of Industrial and Systems Engineering

Spring 2025

Instructions for Assignment Submission:

1. The assignment must be uploaded to Canvas by the indicated due date. The solution should be uploaded to Canvas as a PDF file.
2. Along with your main submission (in PDF format), please ensure you also attach any supplementary files (Excel, Python) that were instrumental in your work. If any Python code was used, upload it in HTML format.

Assignment 1: Linear programming fundamentals

1. **(30 points) Optimal Allocation of Production Resources** A company produces three products, P_1 , P_2 , and P_3 , using three resources: labor, materials, and machine time. The resources available and the requirements for each product are as follows:

Resource	Available Units	P_1 Requirement	P_2 Requirement	P_3 Requirement
Labor (hours)	120	2	3	1
Materials (units)	100	4	2	5
Machine Time (hours)	90	1	2	3

Table 1: Resource Availability and Product Requirements. All requirements are specified per unit of production.

Each unit of P_1 , P_2 , and P_3 yields a profit of \$50, \$40, and \$60, respectively.

- (a) (3 points) Formulate the linear programming (LP) model for the maximization problem. Define all sets, parameters, variables, objective functions, and constraints.
- (b) (3 points) Identify a feasible solution that satisfies all constraints and serves as an initial basis for the simplex method. Explain in words why the optimal solution must lie on an extreme point (and provide a clear definition of extreme points, particularly in this context).
- (c) (3 points) Solve the problem to optimality using revised simplex iterations. Show the calculations for each iteration and identify the optimal solution.
- (d) (3 points) Analyze the sensitivity of the solution by exploring how changes to the resource availability (labor, materials, machine time) affect the optimal solution and the objective function value. In particular, how much should the company pay for an additional unit of each resource (checking each individual resource separately)? For what range of resource levels (checking each individual resource separately) would the obtained basis depict the optimal solution? Show all operations used to solve these questions.
- (e) (3 points) As in the previous point, analyze the sensitivity of the solution by exploring how changes to the profits affect the optimal solution and the objective function value. In particular, how much would the company benefit (or be affected negatively) for an additional unit of profit (for each type of product)? For what range of profit levels (checking each individual product separately) would the obtained basis depict the optimal solution? Show all operations used to solve these questions.
- (f) (3 points) Determine if the optimal solution from the original problem remains feasible or not, as well as optimal or not, when removing resource availability constraints (for each resource). Explain in detail.
- (g) (3 point) Assume that you add a new resource, with available units a , and r_i of resource units required per unit of product P_i , $\forall i \in \{1, 2, 3\}$. Under which circumstances would the optimal solution from the original problem remain feasible? Under which circumstances would the optimal solution from the original problem remain optimal? Write the mathematical expression(s) that you would use (and how you would use it/them) to answer these questions.

- (h) (3 points) Determine if the optimal solution from the original problem remains feasible or not, as well as optimal or not, when removing a type of product (for each product separately). Explain in detail.
- (i) (3 point) Assume that you add a new product P_4 , with required units l_4, m_4 , and t_4 of resource units required (of labor, materials, and machine time, respectively) per unit of product. Under which circumstances would the optimal solution from the original problem remain feasible? Under which circumstances would the optimal solution from the original problem remain optimal? Write the mathematical expression(s) that you would use (and how you would use it/them) to answer these questions.
- (j) (3 points) Write the dual problem associated with this linear programming model. Define the variables, objective function, and constraints for the dual problem. Solve the problem (in Gurobi/Python) to optimality and interpret the meaning of the dual variables in the context of this problem.

2. (50 points) Traffic Flow Optimization Problem

Part 1.

The city's public transportation network is reviewing its transportation system, aiming to improve efficiency and deliver an optimal experience for its customers. The data for this system is provided in the Excel file *Transportation Data*, which includes two sheets:

- The **Travel Time** sheet contains the travel times between each pair of stations in the network. If there is no travel time assigned to a specific pair of stations, it indicates that these stations cannot be directly connected by a single arc.
- The **Origin-Destination (OD) Matrix** sheet specifies the number of passengers traveling from a specific station (row) to a destination station (column).

A subset of stations forms a **corridor** or line when all possible pairs of stations within the subset are connected by direct arcs, forming a fully connected graph. In the provided data, the network consists of three corridors and two **transshipment stations** (72 Street and 100 Street) that allow passengers to transfer between corridors.

The primary goal of this exercise is to minimize the total travel time for passengers traveling between specified OD pairs. For modeling purposes, consider the following information:

- For any user entering or leaving a station, it takes 2 minutes to walk between any stop and the entrance of the station.
- At transshipment stations, it takes 5 minutes for a user to walk between stops belonging to different corridors.
- Any other movement within a single station is assumed to take negligible time.

Your first task is to formulate a linear optimization model to determine the optimal flow of passengers across the network that minimizes total travel time. In your formulation:

- (a) (10 points) Clearly define the sets, parameters, decision variables, constraints, and objective function for this problem.
- (b) (10 points) Implement your model using Gurobi/Python and solve it. Report the optimal objective value obtained from the solution.

Part 2. Designing Routes and Allocating Buses

Building on the transportation network described above, the next part of the problem involves designing routes and assigning buses to serve them. A **route** is defined as a sequence of arcs (direct connections between stations). For this exercise:

- Routes must only consider combinations in a single direction. Specifically, if a route includes an arc (53 Street, SB Park) followed by the arc (SB Park, 72 Street), then the route cannot include any arc that reverses direction, such as (72 Street, 66 Street). This ensures that routes are consistently oriented and do not backtrack.
- A specific route can only include stations that belong to the same corridor. Stops from multiple corridors cannot be combined into a single route.

Using a route incurs a **fixed cost of \$1000** per route, plus an **additional cost of \$100 per station** visited. For example:

- A route visiting two stations incurs a cost of \$1200.
- A route visiting three stations incurs a cost of \$1300.

Each route must be served by buses. Each bus has a capacity of 150 passengers, and the task is to determine how many buses need to be assigned to each route to satisfy passenger demand. Additional constraints for bus allocation are as follows:

- The total number of buses available for the system is limited to 100.
- No more than 5 buses can be assigned to a single route.

Your tasks for this part of the problem are as follows:

- (5 points) Create a graphic representation of the nodes and edges within a station of your choosing, clearly indicate how the connections within the station are structured.
- (5 points) For a corridor of your choosing and considering a single direction, list and illustrate all possible routes that can be created.
- (10 points) Define and formulate a mathematical model to design routes and allocate buses while minimizing the total cost.
- (10 points) Implement your model using Gurobi and solve it. Report the following:
 - The total cost of the solution.
 - Which routes are selected and what stops do they make.
 - The number of buses assigned to each route.

3. (20 points) What a chess-y problem!

- (3 points) You are an avid chess player, and one day you start to wonder what is the maximum number of rooks (and their locations) that could be put in a generalized $N \times N$ chess board, such that no rook is able to attack another rook.
Mathematically formulate this problem, having N (the number of rows/columns in the generalized chess board) as an input parameter. Solve this problem using Excel or Gurobi/Python for $N = 6, 8$, and 10 (show your solutions for $6 \times 6, 8 \times 8$, and 10×10 chess boards).
- (3 points) Now you wonder what would be the maximum number of bishops (and their locations) so that none can attack each other. Mathematically formulate this problem, having N (the number of rows/columns in the generalized chess board) as an input parameter. Solve this problem using Excel or Gurobi/Python for $N = 6, 8$, and 10 (show your solutions for $6 \times 6, 8 \times 8$, and 10×10 chess boards).
- (4 points) Now you wonder what would be the maximum number of queens (and their locations) so that none can attack each other. Mathematically formulate this problem, having N (the number of rows/columns in the generalized chess board) as an input parameter. Solve this problem using Excel or Gurobi/Python for $N = 6, 8$, and 10 (show your solutions for $6 \times 6, 8 \times 8$, and 10×10 chess boards).
- (10 points) Now you wonder what would be the maximum number of chess pieces, using rooks, bishops, and queens (and their locations) so that none can attack each other. Mathematically formulate this problem, having N (the number of rows/columns in the generalized chess board) as an input parameter. Also assume there is a minimum and a maximum of each type of pieces to be used. Solve this problem using Excel or Gurobi/Python for $N = 6, 8$, and 10 (show your solutions for $6 \times 6, 8 \times 8$, and 10×10 chess boards). For the instances to be solved, assume that there should be at least two pieces of each type, and at most 10 of each type. (Hint: notice that if you solve this problem, all the previous three would be special cases of it, so you may want to simply start from here)